Swayam Singh

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EDUCATION

B.Tech (Computer Science and Engineering) — 2020 – 2024 | CGPA: 9.56
University of Allahabad

PROFESSIONAL EXPERIENCE

Research Fellow — 07/2024 – Present
Microsoft Research

- Engineered an **end-to-end code autoformalization pipeline** in Rust, enabling automated translation of natural language and informal code specifications into formally **verifiable representations**.
- Developing **foundational algorithms** that exploit **reinforcement learning gradient optimization dynamics** to improve representation efficiency, stability, and generalization in large-scale models.
- Developed an LLM performing on par with GPT-4o, focused on code generation, instruction-following, and reasoning; worked on post-pretraining and online/offline RL methods.
- Developed a new supervised fine-tuning method, **SeleKT**, that preserves generalization while significantly improving downstream task performance.
- **Led large-scale model training on multi-GPU clusters**, optimizing distributed training workflows and fine-tuning for high-performance code synthesis and execution.

Maintainer at NumPy — 07/2024 – Present
NumPy

- Review, merge, and evaluate core PRs affecting dtype architecture, numerical semantics, and cross-platform behavior.
- Maintain core NumPy with a focus on dtype infrastructure, numerical correctness, and long-term maintenance.
- Work closely with contributors and maintainers across architectures to address portability, performance, and ABI concerns.

Open Source Intern – Numerical Types & Precision (NumPy OSS) — 07/2024 – 09/2024
Quansight

- Led development of **NumPy-QuadDType**, a cross-platform 128-bit floating-point system providing consistent quadruple-precision arithmetic beyond platform-dependent long double.
- Implemented full NumPy dtype ecosystem integration: **casting, ufunc dispatch, scalar types, memory model, and serialization**.
- Built robust multi-platform distribution and testing infrastructure, handling **compiler toolchains, SIMD availability, FMA support, and architecture-specific numerical behavior**.

Machine Learning Engineer Intern — 05/2022 – 10/2022
dataX.ai

- Developed deep-learning models for business-market growth across Computer Vision and language modelling.
- Engineered a comprehensive API to automate conversion of a pretrained model into ONNX format and its seamless deployment via NVIDIA Triton Server, culminating in a 12% reduction in VM load.
- Developed custom CUDA kernels to accelerate 3D image processing for medical scans, achieving a 2x speedup in segmentation tasks over existing GPU-based solutions and improving overall throughput.

RESEARCH AND PUBLICATION

- **Swayam Singh** et al. “**NextCoder: Robust Learning of Diverse Code Edits**” arXiv:2503.03656, 2025.
  - Published in **ICML 2025** main conference | **ICLR 2025 DL4Code** workshop.
- **Swayam Singh**, Niklas Muennighoff, et al. “**OctoPack: Instruction Tuning Code Large Language Models**” arXiv:2308.07124, 2023.
  - **Spotlight (top 5%)** at **ICLR 2024** | **Instruction Workshop** at **NeurIPS 2023**.
- **Swayam Singh**, Harm de Vries, et al. “**StarCoder: may the source be with you!**” arXiv:2305.06161, 2023.
  - Published at **TMLR** (Transactions on Machine Learning Research).

PROJECTS

Virtual Clothing Assistant [PyTorch, Flask, OpenCV] (500+ ⭐ on GitHub)

- Lets users try on different clothes virtually, without going to a trial room.
- Used **ResNet101** and **UNet** to segment the cloth and model respectively, with pose estimation via **OpenPose**; final predictions use the PyTorch implementation of **VITON**.
- Achieved an **SSIM** (Structural Similarity Index Metric) of 0.895.

Numpy-QuadDType
A cross-platform Quad (128-bit) float data-type for NumPy.

- Core developer and maintainer of a cross-platform 128-bit floating-point data type for NumPy.
- Designed a portable alternative to long double, eliminating architecture- and compiler-dependent irregularities.
- Integrated with NumPy’s dtype, casting, and dispatch mechanisms while maintaining performance and correctness.

MIRA – Multimodal Image Reconstruction with Attention

- Designed a transformer-based multimodal architecture for single-view text/image-to-3D reconstruction using ViT encoders and triplane decoders with cross-attention.
- Enabled fast 3D mesh and video generation from a single image in <10 seconds on an A100 GPU, significantly outperforming traditional NeRF pipelines.
- Integrated diffusion-based image generation (SDXL) with structured 3D reconstruction for end-to-end multimodal scene synthesis.

SKILLS

- Systems programming, numerical computing, cross-platform portability.
- C/C++, Python (NumPy internals), memory management, performance optimization.
- Large-scale open-source development, code review, and long-term maintenance.